Programs using control structures Answers

```
1)public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num < 0) {
            System.out.println("Negative number");
        } else if (num == 0) {
            System.out.println("Zero");
        } else {
            System.out.println("Positive number");
        }

        sc.close();
    }
}

2)public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int a, b, c;

        System.out.print("Enter 3 numbers: ");
        a = sc.nextInt();
        b = sc.nextInt();
        c = sc.nextInt();
```

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        if (a > b && a > c) {
            System.out.println("A is greater");
        } else if (b > a && b > c) {
            System.out.println("B is greater");
        } else {
            System.out.println("C is greater");
        }

        sc.close();
    }
}

3)public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int year;

        System.out.print("Enter a year: ");
        year = sc.nextInt();

        if ((year % 4 == 0) && (year % 100 ≠ 0) || (year % 400
== 0)) {
            System.out.println("leap year");
        } else {
            System.out.println("Not a leap year");
        }

        sc.close();
    }
}

4)public class Main {
```

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    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your marks: ");
        int marks = sc.nextInt();

        if (marks ⩾ 90) {
            System.out.println("Grade A");
        } else if (marks ⩾ 75) {
            System.out.println("Grade B");
        } else if (marks ⩾ 60) {
            System.out.println("Grade C");
        } else {
            System.out.println("Fail");
        }

        sc.close();
    }
}

5)public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n;

        System.out.print("Enter the n value: ");
        n = sc.nextInt();

        for (int i = 1; i ⩽ n; i++) {
            System.out.print(i + " ");
        }
```

```
        sc.close();
    }
}

6)public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int num;

        System.out.print("Enter a number: ");
        num = sc.nextInt();

        for (int i = 1; i ⩽ num; i++) {
            System.out.println(i + " * " + num + " = " + (num *
i));
        }

        sc.close();
    }
}

7)public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int num, sum = 0;

        System.out.print("Enter a number: ");
        num = sc.nextInt();

        int i = 1;
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        while (i ⩽ num) {
            sum += i;
            i++;
        }

        System.out.println("Sum of first n numbers: " + sum);

        sc.close();
    }
}

8)public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n, temp, digit, rev = 0;

        System.out.print("Enter a number: ");
        n = sc.nextInt();

        temp = n;

        while (temp ≠ 0) {
            digit = temp % 10;
            rev = (rev * 10) + digit;
            temp = temp / 10;
        }

        System.out.println("Reversed number: " + rev);

        sc.close();
    }
}
```

```
9)public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        for (int i = 2; i < num; i++) {
            if (num % i == 0) {
                System.out.println("Not prime");
                break;
            } else {
                System.out.println("Prime");
                break;
            }
        }

        sc.close();
    }
}

10)import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int a, b, choice;

        System.out.print("Enter first number: ");
        a = sc.nextInt();

        System.out.print("Enter second number: ");
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        b = sc.nextInt();

        System.out.println("1. Addition");
        System.out.println("2. Subtraction");
        System.out.println("3. Multiplication");
        System.out.println("4. Division");

        System.out.print("Enter your choice: ");
        choice = sc.nextInt();

        switch (choice) {
            case 1:
                System.out.println("Addition = " + (a + b));
                break;

            case 2:
                System.out.println("Subtraction = " + (a - b));
                break;

            case 3:
                System.out.println("Multiplication = " + (a *
b));
                break;

            case 4:
                if (b ≠ 0) {
                    System.out.println("Division = " + (a / b));
                } else {
                    System.out.println("Cannot divide by zero");
                }
                break;

            default:
                System.out.println("Invalid choice");
        }
```

```
        sc.close();
    }
}
```